The Geometric Distribution

- Consider an experiment with successive independent trials

- Each trial can result in either Success (S) or Failure (F).

- The probability of success on any one trial is p for some $0 \leq p \leq 1$.

Let X = # trials until the first Success

or

X = # failures before the first Success

Then X has a <u>geometric distribution</u> with parameter p.

Write $X \sim \text{geom}(p)$. (We will write $\text{geom}_1(p)$ versus $\text{geom}_0(p)$.)

Let $X \sim \text{geom}_0(p)$.

$$P(X=0) = P(S \text{ on } 1^{st} \text{ trial}) = p$$

$$P(X=1) = P(F \text{ on } 1^{st} \text{ trial}, S \text{ on } 2^{nd} \text{ trial})$$

$$\overset{\text{indep}}{=\!=} P(F \text{ on } 1^{st} \text{ trial}) \cdot P(S \text{ on } 2^{nd} \text{ trial})$$

$$= (1-p) \cdot p$$

$$P(X=2) = P(F, F, S)$$

$$\overset{\text{indep}}{=\!=} P(F) \cdot P(F) \cdot P(S)$$

$$= (1-p) \cdot (1-p) \cdot p$$

$$= (1-p)^2 \cdot p$$

In general, the <u>probability mass function</u> for $X \sim \text{geom}_0(p)$ is

$$P(X=x) = \begin{cases} (1-p)^x \cdot p, & x = 0,1,2,\ldots \\ 0, & \text{otherwise} \end{cases}$$

The expected value of $X \sim \text{geom}_0(p)$ is

$$E[X] = \sum_{x=0}^{\infty} x \cdot P(X=x)$$

$$= \sum_{x=0}^{\infty} x \cdot (1-p)^x \cdot p$$

$$= p \sum_{x=0}^{\infty} x \cdot q^x \quad \text{where } q = 1-p$$

$$E[X] = pq \sum_{x=0}^{\infty} x \cdot q^{x-1}$$

$$= pq \sum_{x=1}^{\infty} x \cdot q^{x-1}$$

$$= pq \sum_{x=1}^{\infty} \frac{d}{dq} q^{x}$$

$$\boxed{\sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \quad \text{if } |r|<1}$$

$$= pq \frac{d}{dq} \sum_{x=1}^{\infty} q^{x} = \frac{q}{1-q}$$

$$E[X] = pq \frac{d}{dq} \frac{q}{1-q}$$

$$= pq \frac{(1-q)\cdot 1 - q(-1)}{(1-q)^2}$$

$$= \frac{pq}{(1-q)^2} \underset{q=1-p}{=} \boxed{\frac{1-p}{p}}$$

Recurrent and Transient States

- By the Markov property, when started at a recurrent state, the chain will return an infinite number of times.

<u>Question</u>: What is the expected number of returns to a transient state?

<u>Definition</u>: For a Markov chain $\{X_n\}$, denote/define the <u>first hitting time</u> on state i as

$$T_i = \min\{n \geq 0 : X_n = i\}$$

Sometimes we will define

$$T_i = \min\{n \geq 1 : X_n = i\}$$

This is known as a <u>first return time</u> in the <u>context</u> where $X_0 = i$.

Suppose that $X_0 = i$.

Consider $T_i = \min\{n \geq 1 : X_n = i\}$.

Let $g_i = P(T_i < \infty \mid X_0 = i)$

$= P(\text{return to } i \text{ at least once} \mid X_0 = i)$

⊛ State i is recurrent $\iff g_i = 1$.

(State i is transient $\iff g_i < 1$.)

<u>Question</u>: What is the expected number of returns to a transient state?

- $P(\text{return at least once} \mid X_0 = i) = g_i$

- $P(\text{return at least twice} \mid X_0 = i) \overset{\text{M.P.}}{=} g_i \cdot g_i$
$= g_i^2$

- $P(\text{return at least 3 times} \mid X_0 = i) \overset{\text{M.P.}}{=} g_i^3$

Let $N = \#$ returns to state i

We want to find $\underbrace{E[N \mid X_0 = i]}_{\text{Notation: } E_i[N]}$

We can do better!

$$\underbrace{P(N \geq 1 \mid X_0 = i)}_{\text{Notation: } P_i(N \geq 1)} = g_i$$

$$P_i(N \geq 1) = g_i \,, \quad P_i(N \geq 2) = g_i^2 \,, \ldots$$

$$\Rightarrow \; P_i(N = 0) = 1 - P_i(N \geq 1) = 1 - g_i$$

$$P_i(N = 1) = P_i(N \geq 1) - P_i(N \geq 2)$$

$$= g_i - g_i^2 = g_i(1 - g_i)$$

$$P_i(N = 2) = P_i(N \geq 2) - P_i(N \geq 3)$$

$$= g_i^2 - g_i^3 = g_i^2(1 - g_i)$$

In general,

$$P_i(N=j) = \begin{cases} g_i^j(1-g_i), & j = 0, 1, 2, \ldots \\ 0, & \text{otherwise} \end{cases}$$

i.e. $N \mid X_0 = i \sim \text{geom}_0(p = 1 - g_i)$

Thus, $E[N \mid X_0 = i] = \dfrac{1-p}{p} = \boxed{\dfrac{g_i}{1-g_i}}$